No.

W_2:

Bài 1:

$$X_{3\times 4} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

head 0: $d_{k_1} = 2$
$d_q = 4, k_2 = 2$
$\Rightarrow d_{k_2} = 4 - 2 = 2$

Q =

$$W_{q_1} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \quad W_{k_1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad W_{v_1} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$Q_1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$K_1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$V_1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}$$

5 10 15 20 TOTAL

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Bài 2:

Có 2 heads, mỗi head two vector 2 chiều

head 1 $\in \mathbb{R}^{7\times 2}$ head 2 $\in \mathbb{R}^{7\times 2}$

=> concat (head 1, head 2) $\in \mathbb{R}^{7\times 4}$

Số head	$d_k = d_{model} / n_{heads}$
1	128
2	64
4	32
8	16

Bài 3:

pos	PE(pos, 0)	PE(pos, 1)	PE(pos, 2)	PE(pos, 3)
0	0	1	0	1
1	0,84	-0,54	0,01	0,99995
2	0,9093	-0,4161	0,0199	0,9998

Bài 4:

$$h = ReLU(W_1 x + b_1) = ReLU\left(\begin{bmatrix}1\\2\end{bmatrix} \cdot \begin{bmatrix}0\\-1\end{bmatrix}\right) = ReLU\begin{bmatrix}1\\1\end{bmatrix}$$

$$= [1\ 1]$$

Bài 5:

$d_k = d_{model} / n_{heads} = 128 / 4 = 32$

Q/K/V shape: (8, 4, 100, 32) Attention scores shape: (8, 4, 100, 100)

Attention output theo từng head: (8, 4, 100, 32); sau khi ghép heads: (8, 100, 128)

Output sau FNN: (8, 100, 128)

Bài 6:

7	Số phần tử T^2
32	10^{24}
64	40 2x
128	16 384
256	65 536

5 10 15 20 TOTAL

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